

Alejandro Andrés-Juanes

PhD student · Quantum physics

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ISTA, Am Campus 1, 3400 Klosterneuburg, Austria

PhD student in Johannes Fink's group at ISTA researching superconducting circuits, quantum optics, and remote qubit entanglement.

Education

Institute of Science and Technology Austria (ISTA) — PhD student, Physics

2023-10 – present · Klosterneuburg, Austria

Universitat de Barcelona — Master, Quantum Science and Technology

2021-09 – 2022-09 · Barcelona, Spain

Overall score: 92/100.

Universidad de Salamanca — Bachelor, Physics

2017-09 – 2021-06 · Salamanca, Spain

Overall score: 91/100.

Junior year at Universitat de Barcelona, September 2019–June 2020.

Publications

Distributing Stationary Qubit Entanglement through a Nonlocal Squeezed Reservoir

A. Andrés-Juanes, J. Agustí, R. Sett, E. S. Redchenko, L. N. Kapoor, S. Hawaldar, P. Rabl, J. M. Fink

Physical Review X 16, 031005 (2026). DOI: 10.1103/r4jt-j39w

Preprint: Entangling remote qubits through a two-mode squeezed reservoir (arXiv:2510.07139).

Technical Skills

Programming: Python, QUA, MATLAB, Fortran

Nanofabrication: Electron-beam lithography, evaporation, RIE, ALD

Measurement: VNA, digitizer, spectrum analyzer, Quantum Machines OPX

Languages

Spanish: Native

English: Professional proficiency; IELTS 7.5

Catalan: Professional proficiency

French: Intermediate

German: Intermediate

Talks and Schools

Invited talk at SM2Q, Glasgow — November 2025.

Contributed talk at FisMat25, Venice — July 2025.

Talk at APS March Meeting, Anaheim — March 2025.

Quantum Science: Implementations summer school, Benasque — June 2024.

Poster at the 2nd QuantA Workshop — June 2024.

Poster at the SFB BeyondC Autumn Workshop — September 2023.

VCQ Summer School — September 2023.

ICFO Frontiers of Light School on Quantum Materials — June 2022.

QuantuMatter Conference — June 2022.

Awards and Fellowships

Best poster award, Low Temperature Quantum Detectors, Helsinki — August 2025.

ICFO Student Research Fellowship — January 2022.

First prize, Quantum Ideas Factory Hackathon, Heidelberg — November 2021.

EFEQT admitted student — October 2021.

ICCUB María de Maeztu Fellowship — October 2021.

ICFO Summer Research Fellowship — July 2021.

CIC nanoGUNE summer internship — July 2020.

Second prize, National Agroalimentary Olympics of Spain — June 2017.

Bronze medal, National Physics Olympics of Spain — March 2017.

Best overall score award during secondary school — July 2016.

Service and Interests

VCQ student speaker for ISTA — since 2023.

Physics track student representative at ISTA — since 2023.

Cultural promoter and activities organizer, CMU Penyaforat-Montserrat, Barcelona — 2019–2020.

Ten years of clarinet and piano studies at the Professional Conservatory of Salamanca; middle degree in B-flat clarinet.

Experience

Graduate school rotations · ISTA / University of Vienna

2024-01 – 2024-04

Germanium-granular aluminium heterostructures with Georgios Katsaros; non-Hermitian physics with Andreas Nunnenkamp.

Research intern · ISTA — Johannes Fink's group

2022-10 – 2023-09

Circuit design, fabrication, characterization, and measurement for remote superconducting-qubit entanglement through a two-mode squeezed source.

Master's thesis researcher · ICFO — Adrian Bachtold's group

2022-01 – 2022-09

Carbon nanotube double quantum dots for a mechanical qubit; high-inductance superconducting resonator design and RF characterization.

Summer research fellow · ICFO — Frank Koppens' group

2021-07 – 2021-09

CVD graphene growth and transfer, nanofabrication, and Python data analysis.

Bachelor's thesis researcher · Universidad de Salamanca — Nanotech group

2020-01 – 2021-07

TMD-based optoelectronic devices: monolayer characterization with optical microscopy, Raman spectroscopy, and Python image processing.

Summer research intern · CIC nanoGUNE — Nanodevices group

2020-07 – 2020-08

Electrical and Raman characterization of MoS₂ and graphite with organic molecule intercalation.